JEE–MAIN EXAMINATION – JANUARY 2026

(HELD ON WEDNESDAY 21st JANUARY 2026) **TIME : 9:00 AM TO 12:00 NOON**

CHEMISTRY — TEST PAPER WITH ANSWER

SECTION-A

51. Consider the following reactions.

$PbCl_2 + K_2CrO_4 \rightarrow A + 2KCl$
(Hot solution)

$A + NaOH \rightleftharpoons B + Na_2CrO_4$

$PbSO_4 + 4CH_3COONH_4 \rightarrow (NH_4)_2SO_4 + X$

In the above reactions, A, B and X are respectively.

(1) $Na_2[Pb(OH)_2]$, $PbCrO_4$ and $(NH_4)_2[Pb(CH_3COO)_4]$

(2) $PbCrO_4$, $Na_2[Pb(OH)_4]$ and $[Pb(NH_3)_4]SO_4$

(3) $Na_2[Pb(OH)_2]$, $PbCrO_4$ and $[Pb(NH_3)_4]SO_4$

(4) $PbCrO_4$, $Na_2[Pb(OH)_4]$ and $(NH_4)_2[Pb(CH_3COO)_4]$

52. Which of the following represents the correct trend for the mentioned property ?

A. F > P > S > B – Frist Ionization Energy

B. Cl > F > S > P – Electron Affinity

C. K > Al > Mg > B – Metallic character

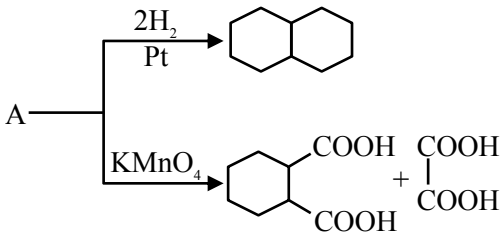
D. $K_2O > Na_2O > MgO > Al_2O_3$ – Basic character

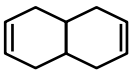
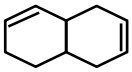
Choose the correct answer from the option given below.

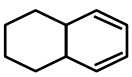
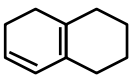
(1) A, B and D only (2) A, B, C and D

(3) A and B only (4) B and C only

53. Identify A in the following reaction.



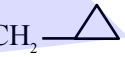
(1)  (2) 

(3)  (4) 

54. A hydrocarbon 'P' (C_4H_8) on reaction with HCl gives an optically active compound 'Q' (C_4H_9Cl) which on reaction with one mole of ammonia gives compound 'R' ($C_4H_{11}N$). 'R' on diazotization followed by hydrolysis gives 'S'. Identify P, Q, R and S.

(1) $P=CH_3–CH_2–CH=CH_2$, $Q=CH_3–CH_2–CH_2–\underset{Cl}{CH_2}$

$R=CH_3–CH_2–CH_2–\underset{CH_3}{NH}$, $S=CH_3–CH_2–\underset{CH_3}{\overset{OH}{CH}}$

(2) P=  , Q=  ,

R=  , S= 

(3) $P=CH_3–CH=CH–CH_3$, $Q=CH_3–CH_2–\underset{Cl}{CH}–CH_3$,

$R=CH_3–CH_2–\underset{NH_2}{CH}–CH_3$, $S=CH_3–CH_2–\underset{OH}{CH}–CH_3$

(4) $P=CH_3–CH=CH–CH_3$, $Q=CH_3–CH_2–CH_2–CH_2–Cl$,

$R=CH_3–CH_2–\underset{NH_2}{CH}–CH_2$, $S=CH_3–CH_2–\underset{\underset{OH}{|}}{CH_2}$

55. Given below are two statements :

Statement I : The number of pairs among [SiO_2, CO_2], [SnO, SnO_2], [PbO, PbO_2] and [GeO, GeO_2], which contain oxides that are both amphoteric is 2.

Statement II : BF_3 is an electron deficient molecule can act as a lewis acid, forms adduct with NH_3 and has a trigonal planar geometry.

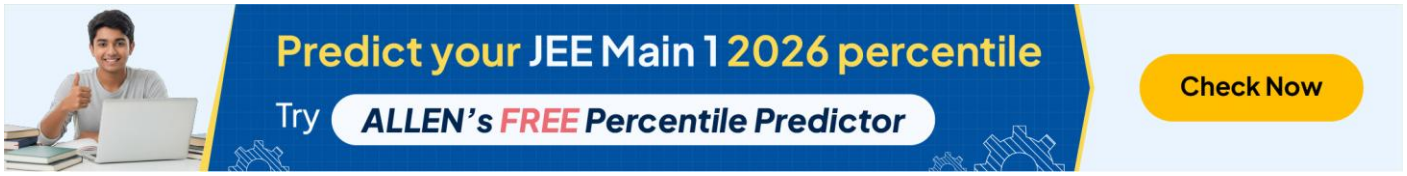
In the light of the above statement, choose the correct answer from the option given below.

(1) Both **Statement I** and **Statement II** are true.

(2) Both **Statement I** and **Statement II** are false.

(3) **Statement I** is true but **Statement II** is false.

(4) **Statement I** is false **Statement II** is true.



56. 80 mL of a hydrocarbon on mixing with 264 mL of oxygen in a closed U-tube undergoes complete combustion. The residual gases after cooling to 273 K occupy 224 mL. When the system is treated with KOH solution, the volume decreases to 64 mL. The formula of the hydrocarbon is :

(1) C_2H_4 (2) C_4H_{10} (3) C_2H_2 (4) C_2H_6

57. 14.0 g of calcium metal is allowed to react with excess HCl at 1.0 atm pressure and 273 K.

Which of the following statements is **incorrect** ?

[Given : Molar mass in g mol^{-1} of Ca–40, Cl–35.5,H–1]

(1) 0.35 mol of H_2 gas is evolved.

(2) 7.84 L of H_2 gas is evolved.

(3) 33.3 of $CaCl_2$ is produced.

(4) The limiting reagent is calcium metal.

58. In Carius method, 0.75 g of an organic compound gave 1.2 g of barium sulphate, find percentage of sulphur (molar mass 32 g mol^{-1}). Molar mass of barium sulphate is 233 g mol^{-1}.

(1) 4.55% (2) 10.30% (3) 21.97% (4) 16.48%

59. Elements P and Q form two types of non-volatile, non-ionizable compounds PQ and PQ_2. When 1 g of PQ is dissolved in 50 g of solvent 'A'. ΔT_b was 1.176 K while when 1 g of PQ_2 is dissolved in 50 g of solvent 'A', ΔT_b was 0.689 K. (K_b of 'A' = 5 K kg mol^{-1}). The molar masses of elements P and Q (in g mol^{-1}) respectively, are :

(1) 70, 110 (2) 65, 145 (3) 60, 25 (4) 25, 60

60. An organic compound (P) on treatment with aqueous ammonia under hot condition forms compound (Q) which on heating with Br_2 and KOH forms compound (R) having molecular formula C_9H_7N. Names of P, Q and R respectively are.

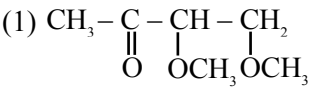
(1) Benzoic acid, benzamide, aniline

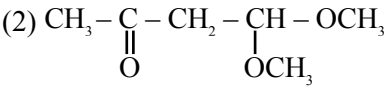
(2) Toluic acid, methylbenzamide, 2-methylaniline

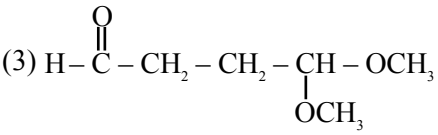
(3) Benzoic acid,4-methylbenzamide,4-methylaniline.

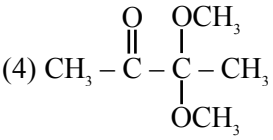
(4) Phenylethanoic acid, phenylethanamide, benzamine

61. An organic compound "P" of molecular formula $C_6H_{12}O_3$ gives positive lodoform test but negative Tollen's test. When "P" is treated with dilute acid, it produces "Q". "Q" gives positive Tollen's test and also iodoform test. The structure of "P" is :

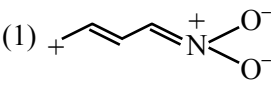
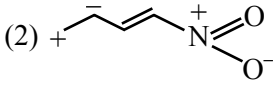
(1) 

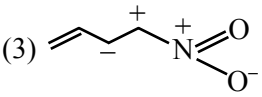
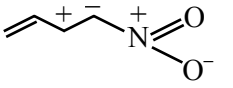
(2) 

(3) 

(4) 

62. From the following, the least stable structure is :

(1)  (2) 

(3)  (4) 

63. MnO_4^{2-}, in acidic medium, disproportionates to :

(1) Mn_2O_7 and MnO_2 (2) MnO_4^- and MnO

(3) MnO_4^- and MnO_2 (4) Mn_2O_7 and MnO

64. Given below are two statements:

Statement I: The number of species among SF_4, NH_4^+, $[NiCl_4]$, XeF_4, $[PtC1_4]^{2-}$, SeF_4 and $[Ni(CN)_4]^{2-}$, that have tetrahedral geometry is 3.

Statement II: In the set $[NO_2, BeH_2, BF_3, AIC1_3]$, all the molecules have incomplete octet around central atom. In the light of the above statements, choose the correct answer from the options given below:

(1) **Statement I** is true but **Statement II** is false

(2) Both **Statement I** and **Statement II** are false

(3) **Statement I** is false but **Statement II** is true

(4) Both **Statement I** and **Statement II** are true



65. Given below are two statements:

Statement I: Among $[Cu(NH_3)^{2+}$, $[Ni(en)_3)]^{2+}$, $[Ni(NH_3)_6]^{2+}$ and $[Mn(H_2O)_6]^{2+}$, $[Mn(H_2O)_6]^{2+}$, has the maximum number of unpaired electrons.

Statement II: The number of pairs among $\{[NiCl_4]^{2-}$, $[Ni(CO)_4]\}$, $\{[NiCl_4]^{2-}$, $[Ni(CN)_4]^{2-}$ and $\{[Ni(CO)_4]$, $[Ni(CN)_4]^{2-}\}$ that contain only diamagnetic species is two.

In the light of the above statements, choose the **correct** answer from the options given below:

(1) **Statement I** is false but **Statement II** is true

(2) Both **Statement I** and **Statement II** are true

(3) Both **Statement I** and **Statement II** are false

(4) **Statement I** is true but **Statement II** is false

66. Identify correct statement from the following :

A. Propanal and propanone are functional isomers.

B. Ethoxyethane and methoxypropane are metamers.

C. But-2-ene shows optical isomerism.

D. But-1-ene and but-2-ene are functional isomers.

E. Pentane and 2, 2-dimethyl propane are chain isomers.

Choose the **correct** answer from the options given below :

(1) B, C and D only (2) A, B and C only

(3) A, B and E only (4) C, D and E only

67. Identify the correct statements.

A. Arginine and Tryptophan are essential amino acids.

B. Histidine does not contain heterocyclic ring in its structure.

C. Proline is a six membered cyclic ring amino acid.

D. Glycine does not have chiral centre.

E. Cysteine has characteristic feature of side chain as MeS-CH_2-CH_2-.

Choose the **correct** answer from the options given below:

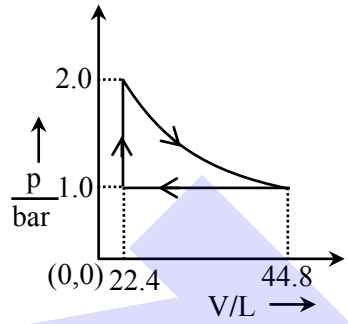
Option

(1) C and E Only (2) B and E Only

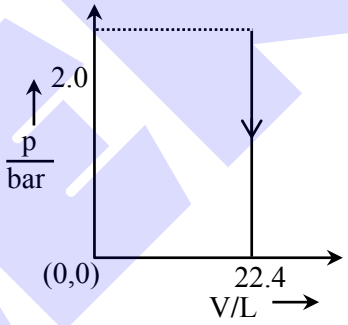
(3) C and D Only (4) A and D Only

68. Which of the following graphs between pressure 'P' versus volume 'V' represent the maximum work done ?

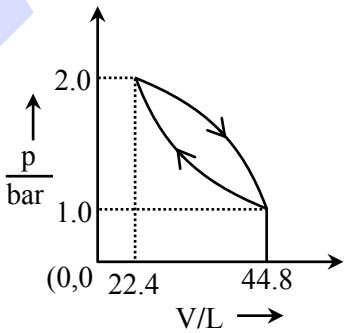
(1)



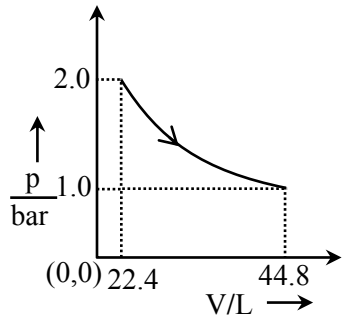
(2)



(3)



(4)



NTA
Allen



69. For the reaction, $N_2O_4 \rightleftharpoons 2NO_2$, graph is plotted as shown below. Identify correct statements.

A. Standard free energy change for the reaction is –5.40 kJ mol^{-1}.

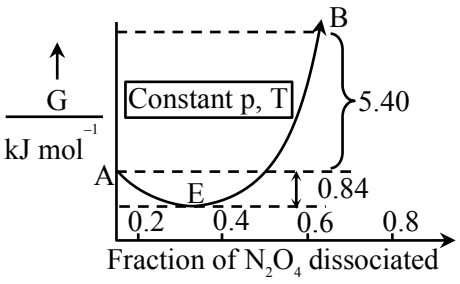
B. As $AG^{\ominus}$ in graph is positive, N_2O_4 will not dissociate into NO_2 at all.

C. Reverse reaction will go to completion.

D. When 1 mole of N_2O_4 changes into equilibrium mixture, value of $\Delta G^{\ominus}$ = –0.84 kJ mol^{-1}

E. When 2 mole of NO_2, changes into equilibrium mixture, $\Delta G^{\ominus}$ for equilibrium mixture is –6.24 kJ mol^{-1}.

E. When 2 mole of NO_2, changes into equilibrium mixture, $\Delta G^{\ominus}$ for equilibrium mixture is –6.24 kJ mol^{-1}.



Choose the **correct** answer from the options given below :

(1) D and E only (2) C and E only

(3) A and D only (4) B and C only

70. Given below are two statements:

Statement I: When an electric discharge is passed through gaseous hydrogen, the hydrogen molecules dissociate and the energetically excited hydrogen atoms produce electromagnetic radiation of discrete frequencies.

Statement II: The frequency of second line of Balmer series obtained from He^+ is equal to that of first line of Lyman series obtained from hydrogen atom.

In the light of the above statements, choose the correct answer from the options given below:

(1) Both **Statement I** and **Statement II** are true

(2) Both **Statement I** and **Statement II** are false

(3) **Statement I** is false but **Statement II** is true

(4) **Statement I** is true but **Statement II** is false

SECTION-B

71. Pre-exponential factors of two different reactions of same order are identical. Let activation energy of first reaction exceeds the activation energy of second reaction by 20 kJ mol^{-1}. If k_1 and k_2 are the rate constants of first and second reaction respectively at 300 K, then In $\frac{k_2}{k_1}$ will be …….. .

(nearest integer) [R=8.3 J K^{-1} mol^{-1}]

72. The pH and conductance of a weak acid (HX) was found to be 5 and 4×10^{-5} S, respectively. The conductance was measured under standard condition using a cell where the electrode plates having a surface area of 1 cm^2 were at a distance of 15 cm apart. The value of the limiting molar conductivity is …….. S m^2 mol^{-1}. (nearest integer)

(Given: degree of dissociation of the weak acid (a) << 1)

73. Use the following data :

Substance	$\frac{\Delta_f H^{\ominus}(500K)}{kJ\,mol^{-1}}$	$\frac{S^{\ominus}(500K)}{J\,K^{-1}\,mol^{-1}}$
AB(g)	32	222
$A_2(g)$	6	146
$B_2(g)$	X	280

One mole each of $A_2(g)$ and $B_2(g)$ are taken in a 1L closed flask and allowed to establish the equilibrium at 500K.

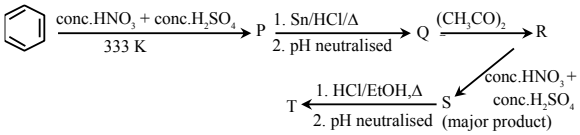
$A_2(g) + B_2(g) \rightleftharpoons 2AB(g)$

The value of x (in kJ mol^{-1}) is …….. (Nearest integer)

(Given: log K=2.2 R=8.3 JK^{-1} mol^{-1})



74. Consider the following reaction sequence



The percentage of nitrogen in product 'T' formed is _____%. (Nearest integer)

(Given molar mass in g mol^{-1} H:1, C:12, N:14, O:16)

75. Consider the following reactions:

$NaCl+K_2Cr_2O_7+H_2SO_4 \rightarrow A+KHSO_4+NaHSO_4+H_2O$

$A + NaOH \rightarrow B + NaCl + H_2O$

$B + H_2SO_4 + H_2O_2 \rightarrow C + Na_2SO_4 + H_2O$

In the product 'C', 'X' is the number of O_2^{2-} units, 'Y' is the total number oxygen atoms present and 'Z' is the oxidation state of Cr. The value of X + Y + Z is _____ .



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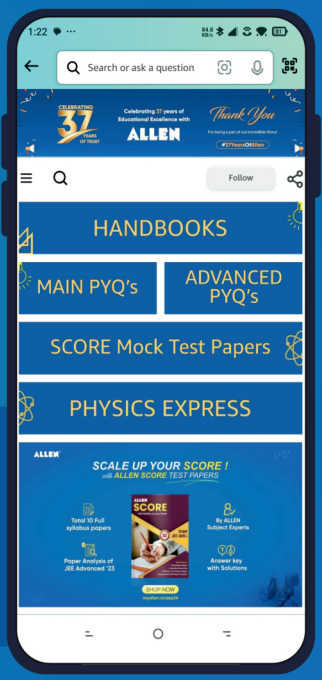
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